OpenClaw AI Script Writing for Peripheral Control Tutorial

> Before starting, please complete the following in the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [05-AI Voice Interaction Configuration], and [06-Three Dialogue Configuration Tests]. If you are currently only using TUI / WhatsApp, you can skip `05` for now. This article assumes the basic environment is already prepared and only covers the AI script generation feature.

1. Tutorial Objective

This article demonstrates through "natural language hardware script generation" that OpenClaw can not only directly control peripherals but also generate reusable script files from your control logic. After completing this tutorial, readers will be able to:

- Understand the basic approach of OpenClaw generating hardware control scripts
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice interaction
- Generate four types of common scripts: RGB, servo, temperature and humidity, OLED

2. Hardware Preparation

- OpenClaw main control board
- Servo / RGB light strip / OLED / DHT temperature and humidity sensor
- `[Optional] AI voice module + speaker`

Description:

- This tutorial does not require connecting all peripherals at once
- Connect the corresponding hardware first depending on which type of script you plan to demonstrate

3. Software Preparation

Code entry points used in this tutorial

- Script generation skill: `/home/jetson/.openclaw/workspace/skills/script-runner/SKILL.md`
- Default script save directory: `/home/jetson/Desktop/`
- Hardware command examples directory: `/home/jetson/hardware_hal_demo/`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

The core of this capability is not "executing an instant command" but having OpenClaw generate a standalone Python script for you. According to the existing skill conventions:

- When the user says "generate script", "write a script", or "save to desktop", it will prioritize using `script-runner`
- Hardware scripts use `openclaw_lib` by default
- The default serial port is `/dev/ttyTHS1`

- The default baud rate is `115200`
- Scripts are typically saved in `/home/jetson/Desktop/`

You can understand this as two usage modes:

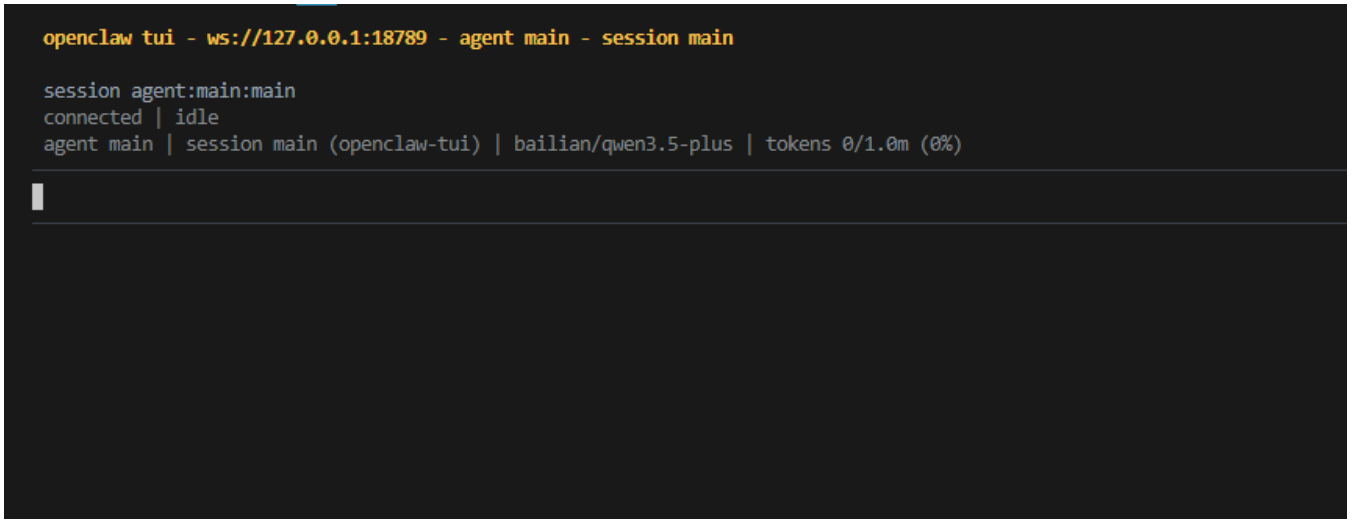
- Mode 1: Generate script and run immediately
- Mode 2: Only generate the script and save it to the desktop for later modification or execution

4. Testing Script Generation Capability

Before starting this tutorial, please complete a basic `openclaw tui` dialogue self-check first; if you haven't done so, see [06-Three Dialogue Configuration Tests]. After confirming that OpenClaw can reply normally, proceed to the script generation-specific test.

Enter the following command in the terminal to enter TUI:

```
openclaw tui
```



It is recommended to start with the simplest script requests, for example:

- Help me write a script to make the RGB red light blink 5 times, save to desktop and run
- Generate a script to read temperature and humidity, don't run it yet
- Write a script to make servo 1 swing back and forth between 0, 90, and 180 degrees
- Generate a script to display "Hello OpenClaw" on the OLED

5. Three Schemes for Communicating with OpenClaw

Scheme	Suitable Scenarios	Advantages	Recommended Use
WhatsApp	Remote requests, classroom demonstration	Can describe the desired script in natural language	Used to demonstrate natural language generation capability
TUI	Local debugging, verify generation results	Easiest to observe script generation and execution output	Preferred debugging scheme

Scheme	Suitable Scenarios	Advantages	Recommended Use
Voice code	Voice interaction, complete AI experience	Can verbally describe script needs	As a complete demonstration scheme

If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue itself does not need to run it again.

- Modify the TTS broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether the replies are broadcast. After modification, save it and run the following program in the terminal
```

- Enter the following in the terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

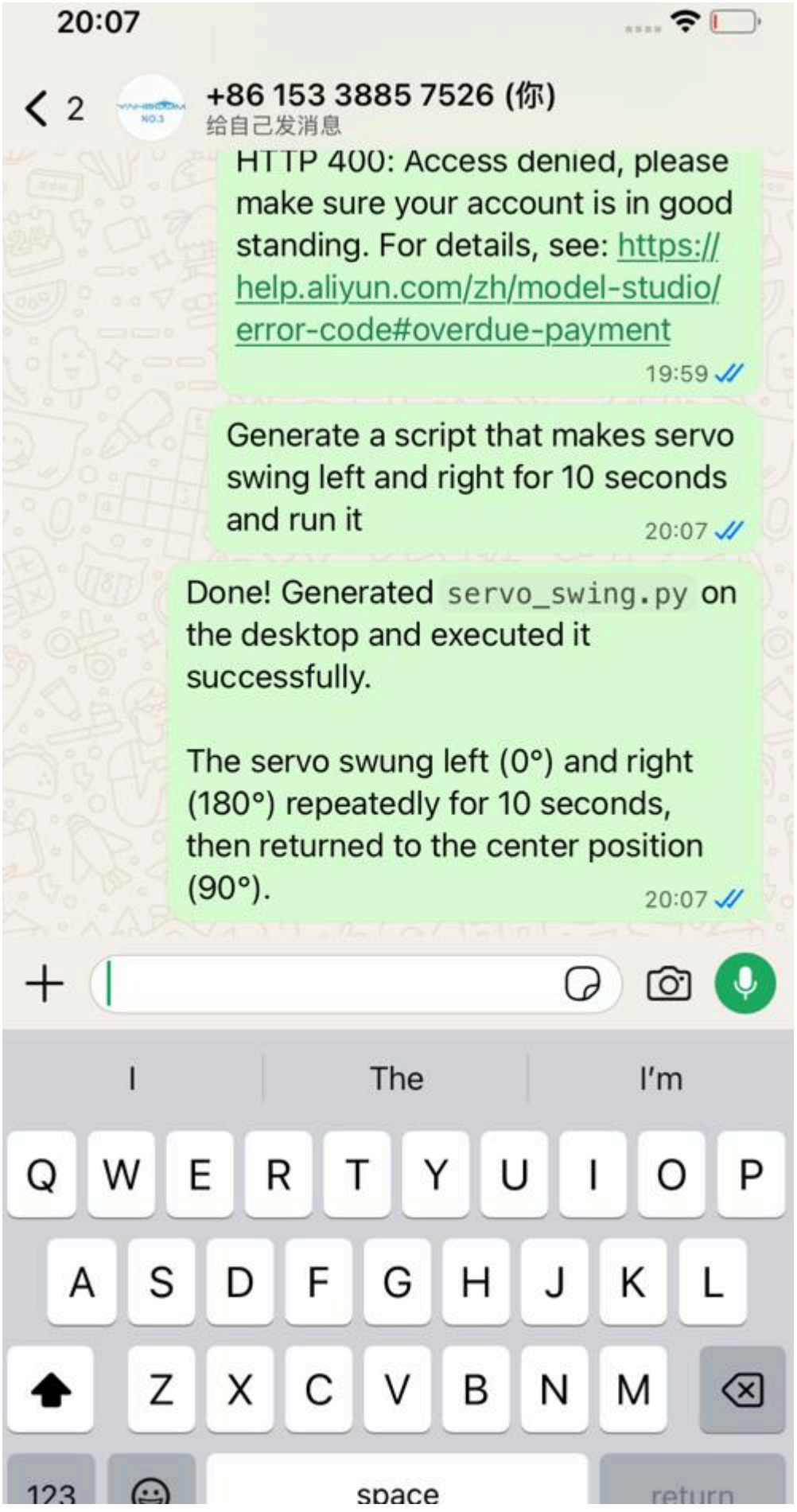


5.1 whatsapp Scheme

For WhatsApp connection steps, please directly refer to [06-Three Dialogue Configuration Tests]. This section only includes dialogue examples suitable for the script generation tutorial.

You can directly converse, for example:

- Generate a script for the servo to swing left and right for 10 seconds and run it


20:07
2
+86 153 3885 7526 (你)
给自己发消息
HTTP 400: Access denied, please make sure your account is in good standing. For details, see: https://help.aliyun.com/zh/model-studio/error-code#overdue-payment
19:59
Generate a script that makes servo swing left and right for 10 seconds and run it
20:07
Done! Generated servo_swing.py on the desktop and executed it successfully.
The servo swung left (0°) and right (180°) repeatedly for 10 seconds, then returned to the center position (90°).
20:07
I
The
I'm
Q W E R T Y U I O P
A S D F G H J K L
Z X C V B N M
123
space
return

5.2 TUI Scheme

If you have completed the TUI self-check in Part 0, simply enter the following command in the terminal to continue the conversation:

```
openclaw tui
```

Then you can directly enter script requests, for example:

- Help me write a script to control RGB to become a blue breathing light and run it



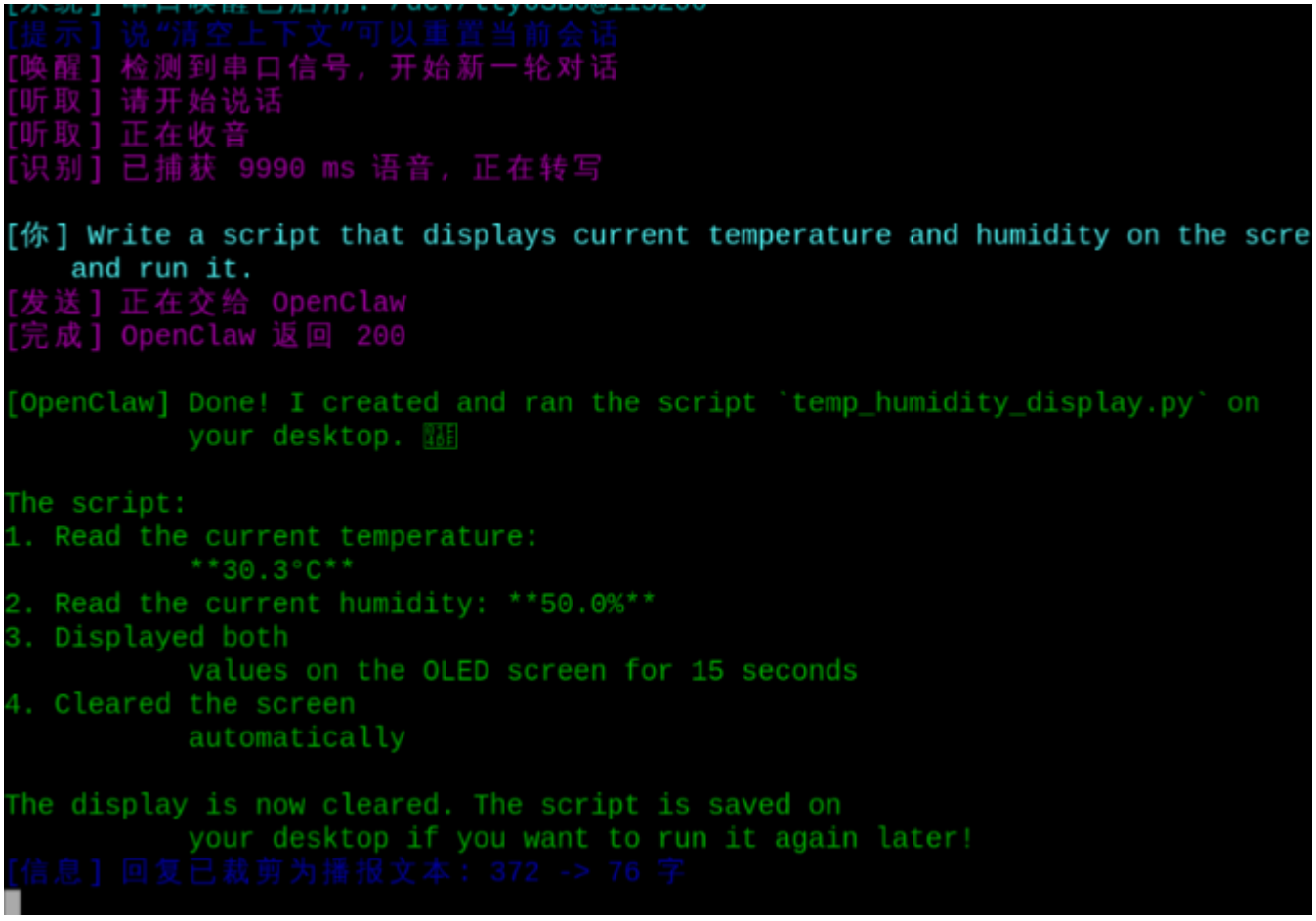
)

5.3 Voice Code Scheme

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [05-AI Voice Interaction Configuration]. After wake-up, you can issue voice commands directly, for example:

- Write a script to display the current temperature and humidity on the screen and run it



6. Comprehensive Cases

Note: The following are all demonstrated using the `openclaw tui` method. You can also choose WhatsApp or voice methods to complete them.

6.1 Case 1: RGB Blink Script

Experimental Objective

Have OpenClaw generate a basic light script to verify the "script creation + peripheral execution" chain.

Effect Description

OpenClaw will organize your natural language requirements into a Python script. After the script is saved to the desktop, you can run it directly or keep it for later modification.

Example Instructions

- Help me write a script to make the RGB red light blink 5 times
- Generate a blue breathing light script, save to desktop and run
- Write a script to make RGB do a rainbow light effect

6.2 Case 2: Servo Action Script

Experimental Objective

Generate a servo action script to turn a one-off conversational action into a reusable script.

Effect Description

This type of script is suitable for actions like "nod", "shake head", "swing back and forth", and "return to center", and is also suitable for further expansion into classroom demonstrations or combined projects.

Example Instructions

- Write a script to make servo 1 rotate back and forth among 0, 90, and 180 degrees
- Generate a servo nodding script
- Help me write a script to make four servos return to 90 degrees together

6.3 Case 3: Temperature and Humidity Reading Script

Experimental Objective

Generate a script that reads the current temperature and humidity and outputs the results.

Effect Description

This type of script is suitable for sensor self-checks, data sampling, or subsequent log recording, and is also the most commonly used basic script type for scheduled tasks later.

Example Instructions

- Generate a script to read temperature and humidity, save to desktop
- Help me write a script to print the current temperature and humidity
- Write a script to read DHT data once and run it

6.4 Case 4: OLED Display Script

Experimental Objective

Generate a script that displays text or metric content on the OLED.

Effect Description

This case is suitable for welcome pages, status pages, and simple dashboards, and is also the foundation for integrating with temperature/humidity sensors, cameras, and other capabilities later.

Example Instructions

- Help me write a script to display "Hello OpenClaw" on the OLED
- Generate a script to display temperature and humidity on the OLED
- Write a script to display a page of simple status information

6.5 Case 5: Combined Script

Experimental Objective

Have OpenClaw generate a comprehensive script that includes multiple peripheral actions at once.

Effect Description

This type of script can chain together "read sensor", "display on OLED", "control RGB", and "drive servo", making it suitable for transitioning from single peripheral control to small project scripts.

Example Instructions

- Write a comprehensive script: read temperature and humidity then display on OLED
- Generate a script: first turn on green light, then rotate servo to 90 degrees
- Help me write a script: if temperature is too high, turn on red light and show alert on OLED